Research-Oriented Problem Statements

Deep Learning for Computer Vision

March 2026

Overview

This document presents diagnostic project ideas for the DLCV course that treat models as *objects of scientific inquiry* rather than black-box tools. Each problem emphasizes **understanding over performance**, uses **pretrained models** to avoid GPU bottlenecks, and produces **interpretation artifacts** (plots, taxonomies, explanations) aligned with CO-4's analysis requirement.

Key Principles:

- One sharp research question per Unit
- Compare paradigms (CNN vs ViT, supervised vs SSL), not individual models
- Minimal training, maximum probing

Requirements:

- Choose one problem statement relevant to your dataset from each of the three Units (You can also create your own similar rigorous problem statements more suited to your dataset and model architecture)

1 Unit I: CNNs and Fundamentals

Problem 1: Receptive Field Archaeology

Question: How does receptive field size affect what CNNs can represent?
Task: Use pretrained CNNs with different depths (ResNet-18, ResNet-50, ResNet-152). Create images with patterns at multiple scales (small textures vs large shapes). Measure where each network's accuracy breaks down and visualize effective receptive fields using gradient-based methods.
Deliverable: Plot of accuracy vs pattern scale + receptive field heatmaps.

Problem 2: Loss Landscape Geometry

Question: Do different architectures have different loss landscape shapes?
Task: Train small CNNs (VGG-style vs ResNet-style) on CIFAR-10 for 10 epochs. Use loss landscape visualization (2D projections along random directions) to compare sharpness. Test: does BatchNorm flatten landscapes?
Deliverable: 3D loss surface plots + written analysis of optimization difficulty.

Problem 3: Data Augmentation as Inductive Bias

Question: Which augmentations actually help, and why?
Task: Train identical CNNs with different augmentation strategies (rotation, crop, color jitter, cutout). Measure accuracy on clean vs corrupted test sets. Hypothesis: augmentation teaches invariances.
Deliverable: Augmentation-performance matrix + failure case analysis.

2 Unit II: Architectures

Problem 4: The Attention Surgery Experiment

Question: What happens when you remove/freeze specific attention heads in ViTs?
Task: Use pretrained ViT. Systematically ablate attention heads (mask them to uniform distribution). Measure which heads are critical for: (a) texture recognition, (b) shape recognition, (c) spatial localization.
Deliverable: Head importance map + examples where specific heads matter.

Problem 5: Hybrid Architecture Trade-offs

Question: What's the optimal CNN-to-Transformer transition point?
Task: Compare CoAtNet-style hybrids with different stage allocations (early conv/late attention vs vice versa). Test on tasks requiring: local detail vs global context. Use frozen pretrained models.
Deliverable: Performance-vs-stage-allocation curves + task-specific recommendations.

Problem 6: Graph Convolutions for Image Superpixels

Question: When do GCNs outperform CNNs on image data?
Task: Convert images to superpixel graphs. Compare GCN vs CNN on: (a) natural images, (b) images with irregular structure (medical, satellite). Hypothesis: GCNs help when spatial grid assumption breaks.
Deliverable: Comparative accuracy table + visualization of when graphs help.

Problem 7: Siamese Network Sensitivity Analysis

Question: What perturbations break similarity learning?
Task: Use pretrained Siamese network (face verification or signature matching). Apply perturbations: rotation, blur, partial occlusion. Measure embedding distance changes and false accept/reject rates.
Deliverable: Perturbation robustness curves + failure taxonomy.

3 Unit III: Learning Paradigms

Problem 8: Self-Supervision vs Supervision Showdown

Question: What does self-supervised learning capture that supervised learning misses?
Task: Compare embeddings from SimCLR (SSL) vs ImageNet-supervised ResNet. Use linear probes to test: texture classification, object counting, geometric reasoning. Hypothesis: SSL captures broader features.
Deliverable: Linear probe accuracy table + t-SNE comparison.

Problem 9: Contrastive Learning's Hard Negatives

Question: How do hard negatives shape contrastive representations?
Task: Use a contrastive model (MoCo/SimCLR). Analyze embedding space: measure distance between (a) same-class pairs, (b) similar-class pairs, (c) dissimilar-class pairs. Test if model struggles with fine-grained distinctions.
Deliverable: Distance distribution plots + confusion analysis.

Problem 10: VAE Latent Space Disentanglement

Question: Are VAE latent dimensions semantically meaningful?
Task: Train VAE on CelebA or MNIST. Traverse individual latent dimensions and measure change in image attributes (e.g., does z[5] control rotation?). Compute disentanglement metrics (SAP, MIG).
Deliverable: Latent traversal visualizations + disentanglement scores.

Problem 11: GAN Mode Collapse Detective

Question: When and why do GANs collapse to limited modes?
Task: Use pretrained GAN (StyleGAN on faces/objects). Sample extensively and cluster outputs. Measure: diversity (FID), coverage (compare to real distribution), failure patterns (memorization?).
Deliverable: Mode coverage analysis + collapse examples.

Problem 12: Diffusion Model Guidance Strength

Question: How does classifier-free guidance affect generation quality vs diversity?
Task: Use Stable Diffusion with varying guidance scales (1.0 to 20.0). Measure: image-text alignment (CLIP score), diversity (LPIPS), artifact frequency. Find optimal guidance for different prompts.
Deliverable: Guidance-quality trade-off curves + prompt-specific recommendations.

Problem 13: Few-Shot Learning Failure Modes

Question: When does few-shot learning break down?
Task: Use MAML or prototypical networks. Test on: (a) high intra-class variance datasets, (b) low inter-class separation, (c) out-of-distribution test classes. Measure where k-shot learning fails.
Deliverable: Failure taxonomy + dataset characteristic analysis.

Problem 14: Transfer Learning Layer Freezing Strategy

Question: Which layers should you freeze when fine-tuning?
Task: Take ImageNet-pretrained ResNet. Fine-tune on small target dataset with different freezing strategies (freeze none, early layers, late layers, all but head). Compare accuracy, training time, overfitting.
Deliverable: Freezing strategy comparison table + recommendations.

Implementation Notes

Resources Needed:

- Pretrained models: Hugging Face, timm, torchvision
- Compute: Google Colab free tier sufficient (inference only)

Grading and Submission

Grading Emphasis:

- 5 marks for interpretation quality (mechanistic explanations)
- 3 marks for experimental rigor (controlled comparisons)
- 2 marks for technical execution (code, plots)

Submission Deadline:

- April 26, 2026
- 4-6 page report focusing on *why*, not just *what*
- Code repository with reproducible experiments
- 10-minute presentation with key insights